

1 Structured summary: | GLS versus Averaged Change-Score
2 Estimators for Clinical Trials with Run-In Observations

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17 *This document is a structured, section-by-section summary of the key points argued in the*
18 *companion manuscript, “| GLS versus Averaged Change-Score Estimators for Clinical Trials*
19 *with Run-In Observations.” It is provided as a supplementary reading aid and does not stand on*
20 *its own; all notation, derivations, simulation detail, and citations are in the main manuscript.*

21

22 **1 Introduction**

23 **The optimality of GLS is contingent on knowing the covariance, which in practice**
24 **must be estimated and may be unstable in small samples.**

- 25 • GLS is the best linear unbiased estimator (BLUE) only when the covariance matrix is
26 known.

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- Feasible GLS substitutes estimated variance components, so its efficiency depends on estimation accuracy.
- Small samples or short follow-up yield unstable estimates that can erode the theoretical advantage of GLS.

The relative merits of change-score and ANCOVA analyses have been studied extensively across several methodological traditions.

- Formal efficiency comparisons for pretest-posttest designs trace back to @brogan1980 and @yang2001.
- @frison1992 and @frison1997 established the efficiency ordering among post-treatment, change-score, and ANCOVA analyses with multiple baseline measurements, the closest antecedent of the present comparison.
- Subsequent work contrasted baseline-as-covariate with baseline-as-response and developed semiparametric covariate adjustment.
- Later contributions clarified when longitudinal models gain power over ANCOVA and adapted these ideas to run-in trials.

The averaged change-score estimator is distinct from and less efficient than the constrained longitudinal data analysis estimator, which is excluded from the present comparison.

- The cLDA estimator places baseline in the response vector under a common-mean constraint.
- It exploits the full covariance structure without specifying the random-effects distribution, ranking between the averaged estimator and GLS.
- A full comparison including cLDA lies beyond the present scope but constitutes a natural extension.

The longpower package implements the GLS baseline but omits the simpler estimators and the misspecification analysis developed here.

- It is the CRAN reference for slope-comparison power calculations under the Liu-Liang, Diggle, and Edland formulations.
- Its routines correspond to the GLS estimator under correct specification, matching the Paper 1 baseline.
- It does not cover ANCOVA, the averaged change-score test, or the efficiency loss from misspecified covariance.

1.1 GLS under the mixed model

The GLS variance for the treatment effect follows from the relevant diagonal element of the inverse information matrix under the marginal covariance.

- The per-pair variance is the treatment-effect element of $(X'\Sigma^{-1}X)^{-1}$.
- The marginal covariance decomposes as residual plus random-effects components.

- The inverse is obtained efficiently via the Woodbury identity.

1.2 ANCOVA with run-in mean as covariate

The ANCOVA variance is governed by the run-in to post-randomization correlation, which determines the magnitude of the variance reduction.

- The factor $(1 - \rho^2)$ scales the variance by the squared marginal correlation between the phase means.
- With no run-in observations the covariate vanishes and ANCOVA reduces to the averaged estimator.
- A large correlation, arising when the random-slope variance dominates the residual, yields substantial variance reduction.

ANCOVA occupies a middle ground by estimating the baseline coefficient from the data while still using the predictive run-in observations.

- Unlike GLS, the coefficient on the run-in mean is estimated empirically rather than derived from assumed variance components.
- Unlike the averaged estimator, it exploits the predictive value of the run-in observations.
- It therefore reduces variance without committing to a full random-effects specification.

1.3 BLUP as theoretical benchmark

The three estimators are linear functions of the change scores and admit a common benchmark in the best linear unbiased predictor of the random slopes.

- Each estimator can be viewed as a linear combination of the observed change scores.
- The BLUP of the subject-specific random slope provides the natural reference predictor.
- It weights the residuals by the inverse marginal covariance using the GLS fixed-effect estimate.

GLS attains the minimum variance among linear unbiased estimators when the covariance is correct, while the simpler estimators are specific approximations to this benchmark.

- The group-specific slope estimators inherit the variance of the treatment-effect fixed effect.
- GLS is optimal among linear unbiased estimators under a correctly specified marginal covariance.
- The averaged and ANCOVA estimators are particular linear approximations to this optimal benchmark.

Misspecification erodes but, in the settings studied here, does not eliminate the GLS efficiency advantage over the simpler approximations.

- GLS propagates errors in the assumed covariance through its inverse weighting.
- The averaged and ANCOVA estimators depend on fewer assumed variance components.

- Section 4 quantifies the resulting erosion; in the misspecifications examined, GLS retains a large margin.

1.4 Misspecified G

Misestimating the random-slope variance forfeits the optimality of GLS while leaving the averaged estimator unchanged.

- An incorrect random-slope variance renders the assumed-covariance GLS no longer minimum-variance.
- The error enters through the inverse-covariance weighting of GLS.
- The averaged change-score estimator's definition does not use this variance component, so its variance does not depend on the assumed value.
- In the sweep below, however, the misspecified GLS variance peaks at 5.30 while the averaged variance on the γ scale is 8.89: the GLS margin is eroded, not eliminated.

1.5 Misspecified R

Misspecifying the residual correlation biases the GLS variance machinery but again leaves the averaged estimator intact.

- Assuming compound symmetry when the truth is AR(1) is a representative misspecification.
- Both the GLS variance formula and the Woodbury correction become biased.
- The averaged estimator's definition does not depend on the assumed correlation, although its γ -scale variance still changes with the true R (8.4 to 9.1 over the plotted range).
- The misspecified GLS sandwich variance stays four- to elevenfold below the averaged variance across the sweep.

1.6 Design

A factorial Monte Carlo design complements the analytical results by varying sample size, design, signal-to-noise, and analysis method under a compound-symmetric truth.

- Sample size and the run-in design configuration are crossed.
- The signal-to-noise ratio is varied; the true correlation is compound symmetric throughout.
- Three analysis methods are compared: correctly specified GLS, misspecified GLS, and the averaged change-score test.

Each configuration is replicated 1000 times and evaluated on empirical variance and power; coverage is not assessed.

- One thousand trials are simulated per configuration.

- Empirical standard deviation and empirical power are recorded for each estimator.
- Coverage of nominal 95% confidence intervals, empirical size at $\Delta = 0$, larger sample sizes, higher ratios, an AR(1) truth arm, and 5000 replicates remain future work.

1.7 Results

The simulation confirms the analytic ordering and quantifies the cost of misspecifying the random-slope variance.

- The empirical standard deviation of the GLS estimator is the smallest of the three under correct specification.
- On the γ scale the averaged estimator is nearly as efficient as GLS in the no-run-in design, but carries two to three times the GLS standard deviation in the run-in designs at $\sigma_b/\sigma = 1$, growing to eight to fifteen times at $\sigma_b/\sigma = 5$.
- Misspecifying the random-slope variance at most roughly doubles the GLS standard deviation, which remains well below that of the averaged estimator in every run-in cell.

2 Discussion

GLS is the efficient choice under a correctly specified, adequately estimated model, which in practice means moderate-to-large samples.

- GLS attains the minimum variance among linear unbiased estimators when the covariance is known.
- Its advantage is realized only when the variance components are estimated accurately, which requires sufficient sample size and follow-up.
- In small samples the instability of the estimated covariance can erase the theoretical gain.

The misspecifications studied here erode the GLS advantage without overturning it; the open robustness question concerns feasible GLS with estimated components.

- GLS weights observations by the assumed inverse covariance, so errors in G or R propagate into its sampling variance.
- In the sweeps of Section 4 the misspecified GLS sandwich variance peaks at 5.30 against a γ -scale averaged-estimator variance of 8.89, and in the simulation the misspecified GLS standard deviation remains well below the averaged standard deviation in every run-in cell.
- Whether feasible GLS, with variance components estimated from small samples, retains this margin is not studied here.

The efficiency gap between GLS and the simpler estimators narrows as the design grows, so the choice matters most in small, information-poor designs.

- With many run-in and treatment observations the phase means become near-sufficient for the random slope.
- The averaged and ANCOVA estimators then recover most of the information GLS extracts, and the relative efficiency approaches one.
- The estimator choice is therefore most consequential when observations are few and the slope-to-noise ratio is high.

Two extensions, the seemingly unrelated regression estimator and the constrained longitudinal data analysis estimator, would complete the efficiency picture.

- The SUR formulation offers a middle ground between the averaged estimator and full GLS but adds implementation complexity.
- The cLDA estimator exploits the full covariance without specifying the random-effects distribution and is expected to rank between ANCOVA and GLS.
- A full comparison including both, and a feasible-GLS simulation under estimated components, is the natural continuation of this work.

3 Data availability statement

The package implementing all three estimators, the simulation drivers, and the shared bibliography are openly available in the project repository.

- The estimator implementations reside in the package source files.
- The simulation drivers and shared compendium bibliography are included.
- Specific paths appear in the Reproducibility section below.

4 Reproducibility

The manuscript was rendered with R 4.4.x using the package source and a small set of standard analysis and build packages.

- The in-package source supplies the estimator implementations.
- The GLS comparator in the simulation is implemented directly in the manuscript source (the analyze functions of the simulation chunk); tinytest is the testing harness.
- The manuscript is built with rmarkdown and bookdown.

The simulation sets a single fixed seed before the replicate loop, which makes the run exactly reproducible.

- A single seed is set once at the entry point of the Monte Carlo comparison.
- No per-replicate reseeding is performed; the replicate stream follows from the single seed.
- Re-running the chunk therefore reproduces the tabulated results exactly.

All code, drivers, companion manuscripts, and the shared bibliography are located at documented paths within the compendium.

- 205 • The package source and simulation drivers reside in their respective directories.
- 206 • The three companion manuscripts are stored alongside this paper in the compendium.
- 207 • The manuscript source and shared bibliography are co-located in the paper directory.